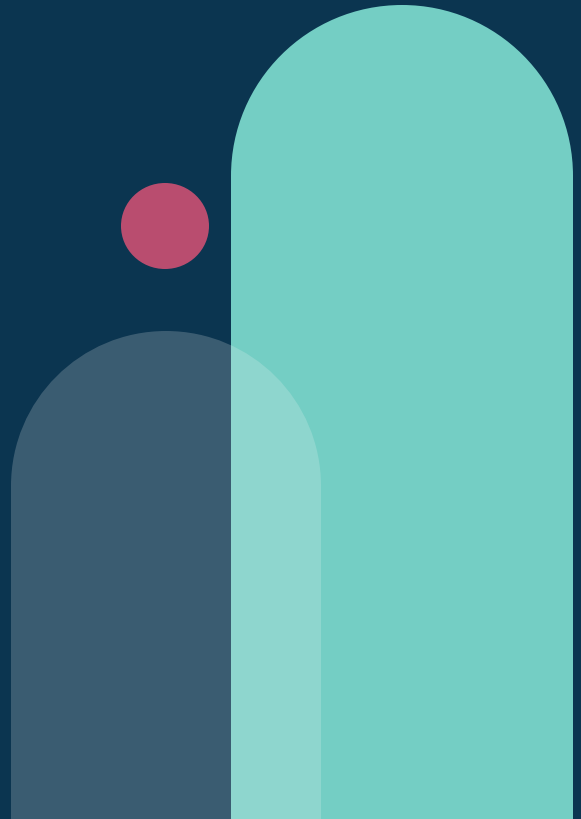
LangChain Fundamentals

01

Structured Output



Structured Output

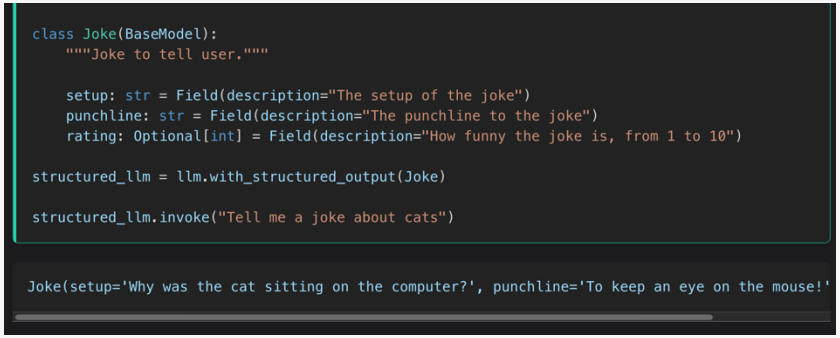
Definition: Structured output refers to constraining the model's responses to a specific format or schema.

Why It Matters:

- Ensures consistency and reliability in outputs, especially important for data processing and storage.
- Makes the outputs easier to parse and use in downstream tasks, like populating databases or integrating with other systems.

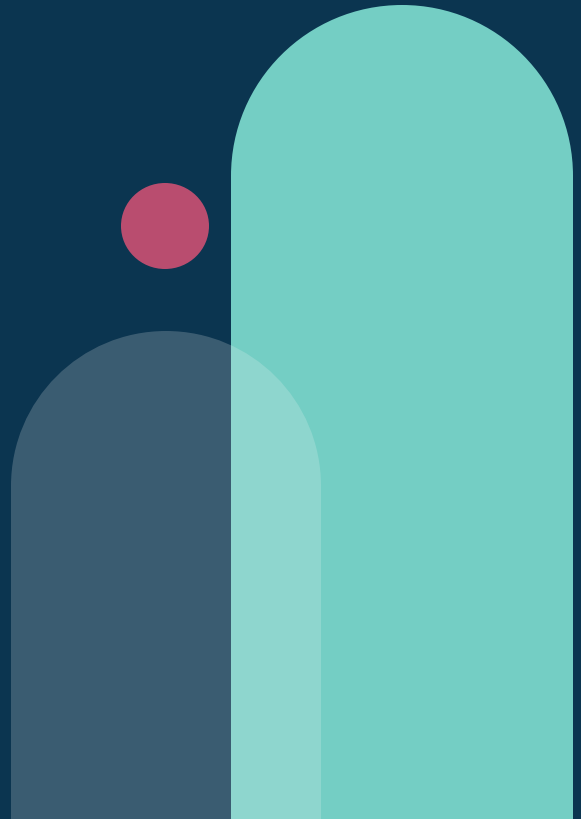
Example Use Case:

- Extracting specific fields from unstructured text and outputting them as PyDantic/JSON objects for further processing.



02

Tool Calling



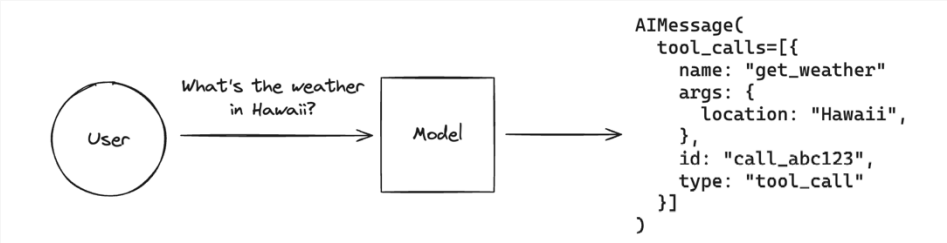
Tool Calling

Definition: Tool calling allows a language model to invoke external tools or functions based on the generated output.

Key Capability:
Precision: The model generates the exact arguments needed for the tool, ensuring that the correct tool is called with the right parameters.

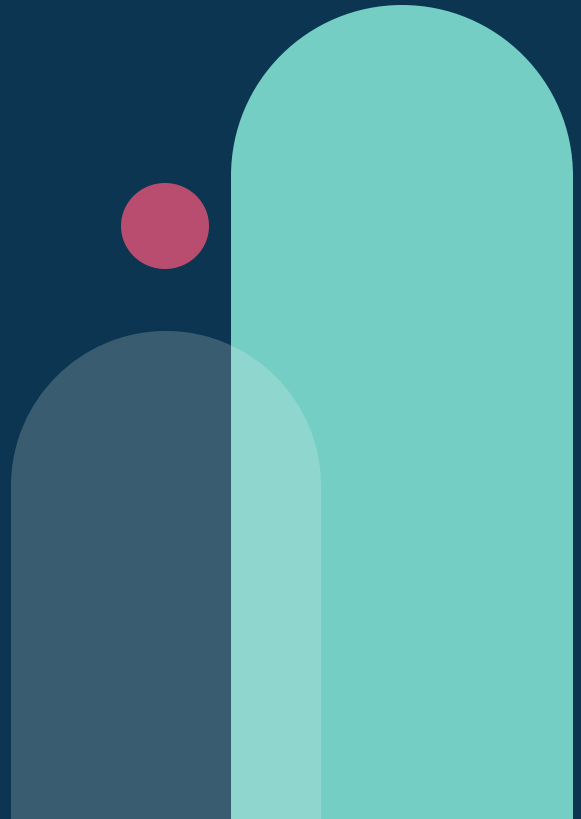
Benefits:
• Extends the model's capabilities beyond text generation to interact with external systems or APIs.
• Useful for tasks like database queries, calling APIs, or performing calculations.

Example Use Case:
• A chatbot that calls a weather API to provide real-time weather updates to users based on their location.



03

Agents



Agents

Main Story: By themselves, language models can't take actions—they just output text. A big use case for LangChain is creating agents.

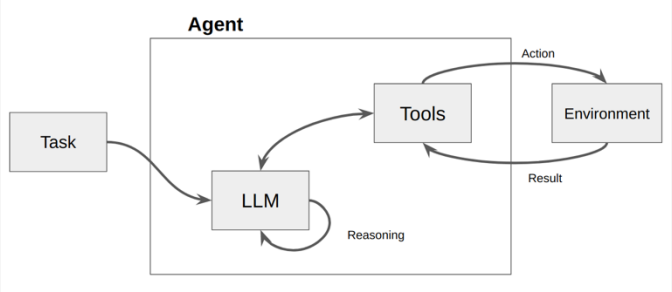


What Are Agents?

- **Definition:** Agents are systems that use LLMs as reasoning engines to determine which actions to take and the inputs to pass to those actions.

How Agents Work:

- **Action Determination:** The LLM analyzes input and decides which actions or tools to invoke.
- **Execution:** Actions are executed based on the LLM's reasoning, and the results are captured.
- **Feedback Loop:** The results are fed back into the LLM to decide if further actions are needed or if the task is complete.



Why Agents Matter:

- **Capability Extension:** Transform LLMs from passive responders to active participants in workflows, making decisions and taking actions autonomously.
- **Example Use Case:** A virtual assistant that autonomously manages tasks like scheduling meetings, sending emails, or retrieving information based on user requests.

Thanks!

Do you have any questions?

